Active InferAnt Stream 016.2 ~ Generative Education: What Will We Do With Our Faculties?

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All right. Welcome everyone. It is December 16th, 2025 and we're live in active inferrant stream 16.2 Two, generative education for today and tomorrow. What will we do with our faculties? This live stream will explore the intersection of human expertise and generative AI in educational content creation, demonstrating how faculty and faculties can leverage cognitive skills and postures and synthetic intelligence tools to create custom modularly configured legibly generated curricula. The presentation will provide open- source resources and methods for people who want to apply these tools today and also hopefully will provide a space for discussion and envisioning the future of generative education and research and what that means for all of us. In the YouTube live chat, please provide any suggestions, questions, comments or prompts. And during the stream we will pick one or more topics to write curriculum for do literature meta analysis on and write a software package and paper. I think we are going to be able to get to all those places. We will look at what is at hand and almost at hand or mandible as it were for three areas of generative education and research and that's going to be curriculum, literature and template. So top level view doxology/curriculum which is the repo I'm in is going to use local LLMs to generate custom course material. So here's the active inference course which we'll take a deeper look at and also we can look at generating introductory biology. So just drag in an HTML file and here we have just a four session intro bio course. So we'll be able to do any number of sessions, any number of modules and any types of materials within the module like lecture lab study note questions, tests, applications, extensions, visualizations with mermaid integrations, further investigations, open questions at the level of one module, one session or we could do it for 50 modules across a 100 sessions or something like that. So that is going to be the curriculum course. And here's what it looks like to run. And we'll start this up. Run pipeline. This is in scripts. Run pipeline. And it will run the test suite. Then it provides some default courses which are in the config courses. So we have active inference 10 modules 20 sessions. Active inference 15 modules 30 sessions. Intro bio two modules four sessions. Intro chem free energy principle 10 modules 20 sessions and intro physics and you can generate all of them in order or you can modify it however you want. So let's let's um await someone's suggestion and then we will modify one of these. So we'll come back to that in a second. The second two repos which I also have open and at the ready are template and literature. So template you might remember from infrarance stream 16.1 about 2 weeks ago. And this is going to be an infrastructure level and a project level for writing a modular manuscript and tested software package that's going to render into all the needed replication unit testing injecting of findings and visualizations into a rendered PDF with citations. So kind of like a power overleaf. Good things about it. You get to write in markdown. You could do entirely prosbased. You don't need any code or any equations at all. Or if you want to use latte and run a ton of code, you can do that as well. So pretty flexible manuscript rendering. And then split off from that is the repo called literature. And what literature is going to do after doing the standard testing and everything, you can search for citations through a variety of literature APIs, download the PDFs, extract their text, and then there's a variety of analyses that are on offer. And where that goes, we get everything from author lists to uh embedding analysis. So using LLM to look at where the papers are in latent space to look at the heat map of similarity across a huge numbers of papers and also look at the principal component analysis different clusters colored by what year. Look at how different language use is associated with different principal components and look at how different papers map out differently in that language space. So supporting rapid configurable andor faculty in the loop literature meta analysis. So, where this is all going and then we'll talk about what hills we're on and what hills we're headed for is kind of a school of tomorrow infrastructural stack. by no means the final, by no means the complete stack, but we have something that with the use of internet APIs as well as local LLMs which compress an enormous amount of background and introductory technical materials, we can develop curriculum research and development all around whatever topic we want. All right. So, Tucker in the chat has suggested the history of China economically and tree grafting from intro to advanced. Let's work with tree grafting and we will start with that curriculum. So, we'll let it chug away in the background while we explore some other topics. Okay. So, I'm in curriculum in scripts run pipeline with Python. And it's not going to matter which course we pick because we're going to overwrite the defaults. So, I'm going to pick one. The course name is going to be tree grafting from introduction to advanced subject expertise area is going to be trees botany garden arburitum science Course level graduate comprehensive language English. We could do this in any language. We'll leave that as default. Number of modules just so we can kind of get through it. We'll do five modules. We'll do four modules with eight sessions. Here we can either just hit enter to go through or you can set a different number of maximum or minimum topics, learning objectives, key concepts, course description. This can be short and again it provides you the default if you want to just hit enter and run through or you can type it in. This course comprehensively covers everything you ever wanted to know and a lot you didn't know you wanted to know about tree grafting. History, biology, sociology, esoterica, deep lore, practical, handson, agro, economic and cultural. All aspects completely covering what the tree gfter of today and tomorrow would be best. apt to know today. Here's additional constraints. This can be skipped or you can say course is for very motivated tree crafters. We have access to internet, all modern technologies and a variety of kinds of trees and people who need tree related services background output default and we'll just clear it. All right. So we are going to in the first step set up the environment and run the test that happens before we configure the course. Then step three is generate the outline of the course. So that's what's happening in this first step. And we can see the logging of the local lama chugging away. Oh, where that should go. I guess the course name what this we'll we'll we'll let that um fix. But it generates a outline and then goes through the outline in generates primary materials which are the lecture and the lab and then generates the secondary materials um uh because okay let's let's do it a little differently. Let's go in config. This will just be a little more reproducible. Write a new course on everything needed to know about tree grafting from basics through advanced and then when we run on main it'll be all properly configured. Um, okay. Meanwhile, let's start up a meta analysis for tree graft related information. So for this one, I'm going to go to literature and I'm going to start a new branch just called live just so that we don't mess too much with stuff. So in literature, we're now on live. So in I'm going to do run literature. We are going to clear the library entirely. Option 7.1 clear. All right, we just have cleared our library. We'll delete the extracted text. Delete the embeddings. Okay, tree grafting. So, here we go. Tree grafting fundamentals to advanced techniques. Now, we will run the pipeline. and get that one started. Okay, that is going to be course and we'll just hit enter to go through all of the defaults. so we've cleared the library from over here. and clear the PDFs. Okay, good. All right. Now, we're going to do meta analysis 3.1. We're going to search just on archive and we're going to turn off the other sources, but there's a bunch of other sources. results per keyword. We'll do 50. Tree grafting. Okay. There was six papers on archive for tree grafting. That was step 3.1. Okay. Move forward to step 4.3. Now we're going to download the

PDFs from archive for the tree grafting papers. Those are going into the PDFs. Let's see how relevant they are. Wow. Semantically informed syntactic machine translation, a tree grafting approach. Here's a syntax paper probably based upon linguistics of grafting. Nested tree recursion post league group algebbras die algebbras grafting operations and Okay, good. So, scalable hierarchical clustering with tree grafting. We will run a meta analysis on those. And so it's going to analyze the papers. So here we have the different language use patterns of these six tree grafting So red means overabundance of word use. Blue is underuse. So PC1 is covering algebra but not tree. PC2 is negatively loaded on tree, positively on algebra and so on. Now what's happening is the Gemma embedding model is being used to analyze the embeddings of these six papers. So, kind of quick recap, when you type in natural language to an LLM, it tokenizes it, calculates the embedding, and then generates tokens from that latent embedding space, pops back out those tokens to you. So, it looks like language in, language out. Embedding is just the first part of that where we're putting in language in this case the extracted text of the paper and getting just the latent space 768 dimensional representation and then that can be analyzed to look a little bit more semantically informed than the PCA. The PCA is looking at word frequencies and then looking at orthogonal principal component vectors that explain or soak up the variance of the variation in language use at a syntactic level. Whereas the embedding space is starting to get us into the semantics of what the topics are about. So even if the word usage were non-over overlapping, for example, different languages discussing the same topic, we could still get the embedding profile. Okay. And looks like in our courses, yep, tree grafting. Here's the tree grafting outline. So that is looking good. Okay. So tree grafting is happening in the background. We're doing the embedding analysis. We're almost done with the embedding analysis on our um tree grafting algebra. And now meanwhile I'm going to start up a new branch on template. New branch is going to be called tree. Now we are going to go to project within template manuscript. So here's the default project and we're going to bring all this in comprehensively. Transform the project to be a variety of useful methods and comprehensively transdisciplinary theoretical and applied review of tree completely update the project. So we get wellformed software package and vast concise manuscript regarding all aspects of tree grafting. We'll use plan mode that. Okay. now let's return to running the full pipeline for the tree grafting. Let it run the test for a few seconds. Course seven. Go through the defaults. Clear. the outline there. [snorts] So, however we'll be most flexible, modular, robust, general. here what should be the primary focus? We want comprehensive review. What should the package comprehensive toolkit? Okay. So now we can look at the embeddings. We can look at the similarity profiles of these different papers. So yellow is the most so on the diagonal. So we see Baker 2014 and 2015. They're the most similar which makes sense. Sounds like it's the first author is the same. And then we have these other dare I say trees dendrograms of the embedding similarity matrix. And we can also look at the raw data of that. Here's a graphical abstract. So this is just sort of yearbyyear metadata keywords keywords changing through time author we only downloaded from archive. principal components and the embedding analysis. So that was the let's let's do a much larger tree grafting analysis. So now we're going to search operation 3.1 and we'll we'll just allow all of them to be used. Now many of them are going to reveal papers that we're not going to be able to get the full text of. Results per keyword tree grafting. We want to fix all the issues. So we are going to transform the template project to be about grafting. Meanwhile, we have a grafting related literature metaanalysis ongoing and we're fixing a little bit the curriculum repo so that we can more easily generate what we already have generated for the other topics. So the plan mode in cursor is uh this kind of little little graphical bug aside is absolutely excellent. So instead of this kind of treadmill issue where you could have an LLM going off the rails and having this kind of eternal sunshine of the spotless semantic space or whatever it is, generating a plan and then repeatedly returning to the plan is a great strategy. So we got curriculum being updated to better generate as that edge case that we discovered transformation of the template repo to be about grafting only and we have completed the search phase. Now we're going to try to download what we have in our bibliography. So 4.3 operation. So now there's 453 tree grafting related papers and we're going to be able to get some of those PDFs like this first one we got from Euro PMC. However, there's going to be other proprietary ones we don't have. Okay. So let's return to the outline here and get through some of these points and topics. Meanwhile, if anyone has any other suggestions and we'll periodically just check back in with all of these different analyses and make sure that they're all on the rails, keep on pushing open source updates, seeing what people are curious about. So, here is where we're at. And to get a better sense of this, let's just look at the introductory biology course materials. So here um an an important feature also is that we can modify the outline before generate. So if we want to look at the outline of curriculum output, we want to go to the biology outline. So if you look at this and you think I think I'd rather just do this for module one. This top level could be written by hand or it could be modified. So that first step gives us a a checkpoint and an opportunity for faculty to to get in the game and our faculties to get in the game for looking at the outline of the course. And then with the primary materials we generate the um let's try this again if we can get to the full We'll delete it seven. Just click through. So it's writing the outline. Let's let's watch that. So um writing the outline is one place where we can intervene where we can look at the outline as written and we can modify the focus or or the extent or just differently design the course. So whether it's a one session educational presentation on a given topic or you want the audience to be different, the language to be different, the points covered, the constraints, whatever you want there. All right, there we go. Now it's going to push that update since now it's working in the generalized case domain. Okay, so now we have this outline of The curriculum for output. Put that guy away for now. Here's the output. course outline tree grafting course outline. So module one grafting theory basic grafting whip and tongue cleft grafting crust bark butdding bridge in arching top working framework troubleshooting and maintenance. So now that is going to generate for each of those modules the primary materials. So it's chugging away here. We're using the Gemma 3 4 billion parameter language model. So you can imagine on a bigger better different computer you could be making these faster or differently. So first it's going to go from the outline and this is happening in SRC are the modular tested methods and then scripts have the orchestrator methods. So three generated outline four generate primary materials generates the lecture lab study notes questions lecture lab study notes diagrams and for the module. So that's now module one is complete. Now it's going to move on to module two and that's going to be chugging away in the background. And then after making the primary materials, it's going to take into the context window all of the primary materials and the course outline and then make secondary materials in the script 5 application, extension, visualization, integration, investigation, etc. So you can you can make other kinds of materials. So that's kind of the different media that you want to project the semantics into. All right. Meanwhile, we have template still chugging away on updating the paper from the example paper into. And let's just run that with option nine just so that it has informiveness and see how that does. Okay, it's still writing all the code, so we won't bug it too much. Um, okay. So, and then at the end of generating all the primary and secondary materials for a course, there's a simple HTML website, including with dark mode for just looking through these materials and um using them. Okay. So, here's the hill we're on. This is where we're at. whether in the chat setting of somebody who's just going to load up a chat GPT or perplexity window, whether they're going to drag in a PDF of a textbook, whether they're going to use some kind of prompt like you're an expert introductory biology instructor, or whether that kind of prompt dispatch is happening under the hood. So whether it's that kind of user interface with a chat or whether it's this more active niche type generation with the custom modular orchestration or whether it's something a little bit in between with cursor with a rag under the hood and all that. We're already in a situation where you can make custom generated curriculum for any topic. And so that's just sort of as if you were to go to perplexity and say make a complete curriculum for basian statistics 10 modules. I'm not logged in on this one. So we'll just see what the free gives you. So whether you just type in here's how many modules I want to learn and here's the topic but however you do it or whatever services come to exist or platforms startups government services etc that offer this we're in a situation where we can generate any number of modules we could do 100 modules on connecting Legos and it will go into the most excruciating fine hairsplitting detail about that. So any number of modules or sessions any combination of educational materials whatever formats we want to project that semantics into customizable and that can be done at the infrastructural level in a way where the whole infrastructure could be all manual. So technically you can have an all manual generation of course materials for any topic any group any language etc or here's the sort of handsoff version and then this script kid approach that I'm taking here gives us intermediate level where we also get interpretability and these intermediate artifacts and so the question is then what and I think Tucker in the chat sort of comes a little bit to this which is we make the curriculum then who does the teaching ultimately and I think that's that's just one of the questions is uh how how is the curriculum experienced and delivered how how is it curated what is brought out how is it evaluated what kinds of domain and higher order skills are being evaluated How do we inform and and support education as a beginning and not not as just this being the end point hardly that's the the open door of the title of what we'll do with our faculties both the organizational chart faculty and our human cognitive faculties and ones that we also don't know about future faculties all of those things are coming into play and we see them chugging away right in front of us. Tucker wrote, "Would be neat if interactive site with homework or practice online worksheets or notebooks?" Yes, definitely. Right now the uh questions part are just static. But that can be intro bio questions. Right now it's just static, but it could be more of an interactive app. Um, okay. So continuing through the outline while we get template is still chugging away. So again this just demonstrates you could look at some of the earlier active inferrant streams where in the early days cursor was a lot more like a musket. It was like prompt it would make one edit and then you would assess prompt edit and then okay now run the test suite. Now prompt here's the error fix that one error. But now this is preparing a plan based upon several And this may be using hundreds or thousands of tools under the hood. And that's the importance of starting in such a heavily scaffolded niche. dare I say grafting into such a niche because we already have all the infrastructure at the top level to render the manuscript and the style constraint like hey we use thin orchestrator scripts to call fully tested documented methods which is now replacing the the previous figures orchestration script with the new methods that are all been flipped. flipped out with this whole SRC folder with all this graph data. So that that still chugs away. Looks good. Now we have this 10 module course. We're 70% done with the primary materials. So again, we'll just let them both chug away a little bit. see how perplexity did here. So here's that basian probability question and it'll say write module one fully all the lecture lab questions. So it's a little bit like the chat version of what here is happening in the sort of Linux file system nested folder structure style by iterating over folders and doing it all in this versionable space where we can track it with git line by line. And that's a great intermediate stop because it plays nicely with operating systems and git. And then you can imagine that and there already are more advanced ways where you have databases and that gives a little bit of a more scalable strategy to versioning material having metadata with every single row and cell merging all these different kinds of things that would be a little bit of a hassle to do in the file structure setting. It's possible to do that in a in a more databaseoriented way and thankfully that's also easy to flip over. So just update this from using file structures into using this database strategy. I try to show mainly at the file and the folder level partially because it's uh again fun investigatable interpretable at hand and helps bridge the continuity between these synthetic systems and augmented coding and just good old revealing in the finder and just seeing okay wait these are just plain texts that are being called by other programs. So here's the module one and again this is already a level of educational accessibility that's just pending somebody typing in the questions or even I want to learn write a course I do not know where or how to start. So that to me is the sort of chat of what is possible with an inkling and an agency to learn. Here's the modules. That was a fresh chat. Maybe it's tracking cookies and modules and stuff like that. Maybe not. And here even we we see a slightly different kind of modules, but nonetheless. So whether again it's that chat format or whether we're in this augmented multiodule investigatable curriculum setting that's where we're at. Okay. So while these are still okay now it's rewriting the manuscript. So that is going so it wrote the code it updated the src folder first and the scripts. So src are the tested methods scripts are the orchestrators of those methods within the project which is now grafting. Now it's going section by section through the manuscript files and rewriting them within this constrained with the cursor rules reminding it of how to actually do citations and all of that to update the manuscript. And once that happens, we should be able to regenerate the PDF. So there's the tests. We'll have that update all the project related tests and documentation to pass. So now we have two kind of cursor sub aents going at once. Also, this is a relatively newish version where you can see diff different just another view on the IDE editor. Okay. Kind of lost the regular version. There we go. B control B. All right. So, we have project tests getting fixed, manuscript getting written, and we'll be ready to run it when it fixes. All right. So in terms of tools right now I'm using cursor 2.2 let's confirm exactly 2.2.23 on Mac OSx on this machine. So each of these three repos, the literature meta analysis, the project template and the curriculum development, they all were made over the last and versioned over the last several weeks using this sort of stack with cursor and using Olama for local LLM calls. All of the the cursor LLM calls are to the cloud and the local calls are to mostly the Gemma 4 billion parameter model, but I tried a few other ones just to explore. Each of the repos has a readme. They're all within the extent of what is reasonable aiming to be modular configurable and to have the local LLM calls with Olama. Uh we are generating custom curriculum. Yes, we are now finishing up. Now we're into the generation of secondary materials. So we finished the primary materials. There's processing all. So that finished primary. Now it moves into generating secondary. So now we're getting the extensions. So that is going to take in all of the primary materials from module one as context and then write several extensions. So now we're generating secondary materials. Um we did a live demonstration of the course outline generation that was right here. There's the tree grafting outline. We could have modified that to to change the focus. Then we can generate different content for different educational contexts. And I I don't even know if balancing is the right metaphor, but something like that, blending, merging, fusing, surmounting this possibility to generate and the the knowhow and know that and wisdom to know the direct consequences of manual and dispatched educational activities. ities and just where, how, when, why, where, all that we blend. And how do we certify and and mark the epistemic status of different kinds of materials? And this is not really covered in the current um packages other than just that you can track line by line where everything is coming from. But how how do you assess and validate that something has a a primary factual accuracy? um a as well as further topics. Testing still being improved. Manuscript almost completed. So, we should get to that manuscript pretty soon. Okay. So, for for those watching, definitely write any comments or questions or we can pick a different uh topic. to explore central question. What will we do with our faculty? Whether it's active inference institute faculty or college of the redwoods or school of tomorrow or whatever you are faculty of or at.

And of course the double triple play with our own faculties they need to be developed within and across generations. So what will we do with our faculties today and where and what will our faculty be tomorrow? What is the role of human in AI assisted education? What is the role of AI in human assisted education? What is the role of the human in human education? What is the role of the AI in AI education? Where and how do we have different standards and quality in spirit and in letter in fact? And h how do we signal that higher order ontology to have interoperability even within a given educational institution when there can be such a a tremendous variety of of formats? And what what are these different roles that that faculty might be in? I'm very excited to see how the manuscript goes, but while it's chugging, I'm not going to interrupt. A few different topics just that I wrote down manually. while I knew that generating things would be happening. So, Taylor series and generalized coordinates that'll be familiar if you've taken the textbook group. Taylor series being a way instead of doing explicit prediction out into future time steps you take a a series sum over higher and higher polomial orders. So here the underlying functions a sine wave we have a first zeroth order approximation just the value first order slope second order quadratic and so on. um and the the tailor series and the very related generalized coordinates higher moments of statistical distributions are where we see a harmonic convergence of multiscale tactics and strategy as a nested continuum. So that's not a situation whereas in the discrete modeling nested literature we might have like discrete modeling at the level of century and then 10 steps within that the level of decade and 10 within that at the year 365 within that for the the day and that would be the all discrete kind of like a piano with finer and finer keys more and more explicit rollouts kind of like a chess algorithm but when we're in the continuous state space we see this all at oness of strategy and tactics. So how do we raise that and build that awareness to transfer at the meta and at the domain level that multi-scale simultaneous tactical strategy trim tab steady cybernetic hand as we move forward and as students move forward in their own educations and ways. Um tests are passing. Let's run the rendering. We will do um add coverage to get to 90% testing successfully. ensure we get all the way on through rendering the whole manuscript. We'll let that one plan out first. Okay. Meanwhile, we are now deep into So, six doesn't have secondary materials, but five we do see the secondary materials. So, that's still chugging away locally. So now we're about 3/4 done generating this course. There is a recursive and quantum cognitive effect that arises through learning and applying active inference. Not saying that that's the only or the best way to even point at that moon. However, when we all of a sudden have these different domains that we can learn from at the ready and it's sort of like well they're all situationally relevant and important. So what to learn and and how do we pick what to learn? There's this really interesting and helpful nomadic reality for learning about active inference, free energy principle, synthetic intelligence, cognitive security, all these kinds of topics. It helps us in our own navigation of this space. And that's this sort of recursive layer that can be approached from analytical, computational, empirical, philosophical, phenomenological, all these different ways of analyzing it. And then it's like, oh yeah, we could project out the train semantic space into a more philosophy oriented active inference course. or we could transpose this course into another language or perspective for sure. That's cool and important even with a classical introbiology course. And I guess that's my own statistical bias uh preference that there's something very interesting that I that I look for to other people's kind of ways of working through that that when it comes to studying sort of the the nut of the issue, the kernel of perception and action and different ways to study perception and action. different frameworks for comparing different kinds of inert on throughactive systems in this cyberphysical setting where we have new flavors of inert on throughactive. For example, if we have a text file within a repo, it has a causal influence through cursors rag even though it's not a program, but still having a given piece of semantics through the soft layer of the LLM comes to play a causal role. comes to be a difference that makes a difference even if it's tucked away in a text file somewhere which is not exactly simply the Turing vonoyman type operational difference and so then it kind of leads one to think about a post lisp in a way where we're working in these repositories that that blur the line in this sort of Mcsher Goodell Cherbach style with what is active and passive, what is reflexive on what. We can have Olam, the local LLM, reviewing its own code and doing different kinds of projections and and mirroring of its own code. So that kind of pattern intuition pump nexus can apply to quote individual which is to say top down or seeing the many as one or the high road as well as a group level collective intelligence bottom up low road. And that is right at the heart of you know basically is the ant colony the organism or is the ant colony a superorganism and how do we talk about these multiscale systems through through spaces and times different perspectives of collective systems which all are. And then how do we deal with that from the high road and the low road? And how does active inference help scaffold that whole process? And as alluded to a few minutes ago, debatably and engagingly these kinds of topics and studying the the essence of cognition itself in the cyberphysical setting applies and can be applied relates to etc. analytical axiomatic ways of thinking to empirical evidential ways of thinking and then also to the phenomenological interiority view from the inside metaphysics.

we're in this augmented cognitive niche where even just simplifying down to me and my body and this repo as a as a external niche to the body. We're going from the body keeps the score to the body and the GitHub repo keep the score on through the niche is not just keeping the score. It's engaging in these extremely specific. It's not about more or less advanced or sophisticated as if there could be just one measure of that. Like who's to say that this is more sophisticated without defining what that means than a clam? Um, but a lot is going on under the hood. So now we're all keeping the score. Everything's keeping the score. Everything's in the game. We're in this massively intraactive setting. And what does that what does that in general in principle help us reduce our uncertainty about? Well, not too much more thankfully than the posture. It's like saying in the the basketball triple threat, pass, dribble, shoot. You have that posture because there's degrees of freedom and there's variability of settings and context. it's like not resolvable or is resolvable that it isn't resolvable which one of those three options is best because there's locations and settings where even in the same position different people would have different suggestions. So, cognitive science helps bring us into this posture where expertise can be expressed rather than something like a far farmer's almanac of just giving us the information on which educational materials to create. That I don't think will ever be the case because there's this open-ended non-stationerity of the educational setting and niches, requirements, all of that. But we're getting to being able to assemble the Megatron in the triple threat at 100,000x speed and what is decided upon what we actually present for that tree grafting course. How much more manual work do we want to put in beyond the generation of these modules? Th those are resolved as unresolvable. So those are those are the current hills is how we reckon with these available local LLMdriven opportunities. Let's see [snorts] if we can get at least a preview of the graft paper. still failing a few tests. So, how do we how do we look from those hills that we're working on right now that that are available again whether to the student or educator andor educator who has access to a chat session with an LLM on through those who are delving deeper into the infrastructure to use these repos or other analogous repos. that make uh more of an industrialized commoditized approach. Um the next hills have as many ought as what ought we do what should we do with these capabilities as they have is questions. What what is the case? What would support active, effective, accurate, aligned generative synthetic intelligence? These are technical and objectively subjective questions. Whereas the is ought gap remains undefeated and even having the confidence like yes we think that we can generate from this configuration with this language model hands off or with these expert educators reviewing it whatever the case may be. We feel confident that we can make accurate, complete, compliant introductory biology curriculum. So great, how do we go from that is at the n equals 1 or at the n equals 100,000 into what we ought to do, how we ought to proceed with our tree grafting education. Okay. So that completed. Now we have we have the website for tree grafting education. Let's look a little bit. So HTML just drag it on. So here's the lecture key topics lab worksheets study notes key concepts application citrus disease extensions Epigenetics, microbiome, 3D bioprinting visualization autogenerated plain text rendered mermaid. Whoa. integration. So, here's connecting some dots with different with the other modules kind of giving a preview and a review investigations. How how do investigations happen? like introbiology, those findings, those claims in the textbook can be traced back in in many to most or all situations to primary experiments done at some prior point andor ones that can be done right there in the lab. So connecting the research and inquiry part first principles research and inquiry like the lab section of a bioclass connecting that to the literature meta analysis that's why that gets kind of partitioned over to the literature repo some of there's some slight differences in mermaid syntax between rendering it on GitHub versus the website But um it's it's out there. Um so that was the Let's make a short course for active inference and AI. So, not that it will hit the grand slam on the first try, but to give a starting point for for people to even who would say, "Oh, no way. I would do it totally totally differently." It's like, "Great. Well, put what you want to put on the table or work with the clay that's on there." So, let's do write another course. It's going to be called active inference AI short. should cover in three sessions total. Three modules. all the basics and intuitive and important relevant features of active inference in relationship to 2025 slash today's generative AI landscape topics also like alignment spatial web multi- aent emerging applications etc. Meanwhile, we will render R. Get us up to 90% plus coverage in tests. then that's a little bit of a strict um cut off to have 90% test coverage before even proceeding with the PDF rendering. But that's just so that it's faster and clearer when something is failing lest we go off really far down a certain road that's that's untested. But that could be relaxed. Meanwhile, we will um keep the 90% and just have it add the tests. All right. Here we will clear scripts. Run the pipeline. All right. This is active inference for generative AI. So that's course two. So I just typed in two. Now we could modify the uh here's here's what the config file looks like. We could modify the name, subject, description, number of courses, etc. or just hit enter just to go all the way through. All right, while that is chugging away, let's see if we have gotten to where we need here. Okay, 89% very close. Very close. in arching. Okay. Previous sessions explored traditional grafting technique whip and tongue cleft and saddle graft. Today we're shifting our focus to a distinctively distinctly different yet increasingly valuable method inarchching. In arching derived from the Japanese term meaning to weave together represents a grafting technique where a scion is attached directly to a side shoot of the rootstock. Let's see if how well we would do on the questions. Okay, questions might have a little bit of context rot, but that's the context engineering question. making sure that everything's generated from from the right um place with the right context in there. Um okay, here we have the short course is already chugging away. Let's look at the outline. curriculum output AI short modules up outline. you know how many times over the years since since each one of us probably got in the game where's the basic course on active inference or how does this apply to generative AI? Not that this is again cannot be emphasized enough that is where the faculty come into play. The faculties come into play is in either knowing to prefer not to even look and give a fresh take on presenting out what what the person thinks uh such a course should be. run to the PDF see if it goes now but here's just a starting point and then this outline we could have intervened at that step to to modify so introduction to active inference active inference and generative AI advanced applications in future directions. Multi-agent, shared world models, precision waiting, grounding language, and action. So may maybe somebody thinks that's the right outline given the the context and and the group that they're teaching in that moment. Maybe it's totally off base, but that's what ought to be taught, not what is being generated right here. Oh, 8939. So close. Um, update to add more tests and relax the requirements to 80%. so now here's the modules. Just push an update in the So here we are. module one. There's the lecture lab. Okay. So, we're going to manipulate a visual stimulus to systematically increase or decrease the perceived surprise relative to location. Record changes in the reported part per perception. Quantify the relationship between perceived surprise and corresponding perceptual adjustment. Develop an intuitive understanding of how action reduces variational free energy. Document the observations to illustrate a predictive coding loop. Participants are seated comfortably in front of a computer screen. software's launch dot stimulus is displayed without any intervention. Participants are asked to report subjective perception of the dot's horizontal position left right on a scale of -3 to plus three. Then the software moves the dot by one degree per second. Participants continue to report subjective perception of the horizontal position at 30 second intervals. then returned to center conduct debriefing. How did the participants reported perception of the dot's position change when the dot was moving to the right? What does that suggest about their predictive model? perception of a dot change when the dot was returning?

Yeah. Funny. Little bit of mixed up with some basic biology questions, but the other questions are accurate. See if that will pass through now. All right. Now we're getting further. Now we're failing because we're hitting some missing figures. Fix all missing figures and needed references to fully render the grafting project and paper. Okay. So we have curriculum using local LLM to generate now the getting into the secondary materials for this three module active inference generative AI course. We have cursor fixing up the tree grafting paper which we should be able to get rendering Then forgotten but not forgotten. We have the attempt to download. We'll just 167 of the PDFs. We'll stop that and we'll just run the literature analysis. We'll just do 6.1. So it's going to do literature meta analysis without any embeddings cuz Olama is busy with the other curriculum repo. So here we go. So now we have meta analysis 459 papers that just happened. Here's publications by year tree Here's how many PDFs we have. We can look at what what completeness we have across different metadata. Keywords grafting trees grafted roottock growth fruit keyword evolution AJ tree maybe Dr. tree was writing about grafting different venue ciiaiculture act bone grafting so PCA analysis we can see there's two clusters it looks like two main clusters of literature got green cluster up here got a blue cluster and then a few down here So we have basically blue and the green have similar values or you know roughly similar on PC1 but the green cluster has high on PC2 language use. Here's the 3D principal component. We can kind of see that there's two separate clusters here. Purple cluster and a green one. So PC1 is just loaded up heavily on grafting. PC2 is negatively loaded on grafting, but it's positively loaded on bone and bone PC3 is positively loaded on artery. PC4 is loaded up on tree. So you can see that a paper that's discussing bone grafting would be positive on PC 1 and two artery grafting on one and three. And we can sort of look at the language use a little bit more resolution to look at what kinds of words are positively or negatively associated with each of those principal components that are calculated from the papers. So the principal component one is talking about bone grafting not about rootstock citrus apple orange but PC etc. So, here's bone grafting is this vector pointing up and to the right. Grafting in general down into the right. And then here's trees coming off the left. Cool. Very cool. So, I'll just push that to the live branch. old literature, but that didn't affect the main. back to template. Um, okay. Let's look at the website for the active inference and then see if we get All right, now we're getting the grafting simulations are happening. Let's see if it will render us the PDF. Great. Okay, looks like we should see the manuscript very soon. So that's again using the within the project. So the infrastructure top level never changes but infrastructure methods are getting called here to run the test suite, run the figure generation and render the manuscript. So here's all the modular sections about the tree And then the output is going to go into PDF. This is rendering the markdowns one by one. And then it's going to output a combined up. It's still hashing out some unresolved. We We will stick around to to get to render this PDF. Um so showing it can take a little bit more than an hour for all of the steps to to really fully gel. I think that can be improved uh as well. It's just showing what's out there right now. Then we'll return to the active inference. Okay, we have missing figure. So, there's that Okay, cool. Let's see if this will render the PDF. Then it does uh uses the local LLM to review. Let's just see if Oh, okay. Here we go. So, title needs to be changed. So we will do for the project we have give a informative title. Right now it is update the subtitle author is inferant number 016. We'll see if there's any other changes to make. Then clickable clickable PDF software package grafting anatomy equations methods. These citations might be hallucinated. We'll just we'll check this one for example. But that's why there's the literature repo. that's a real one. That's a real citation. But with the with the literature repo running then you would know that it was real whatever that's worth. So here we have methodology here's more equations related to Okay species compatibility matrix. So I injected the heat map in temperature effects, humidity effects, discussion, cultural conclusion, future acknowledgements, detailed economic parameters, extended techniques, supplemental results, biogenetic analysis. make any other updates to improve the I'll just dump in all the manuscript sections. kind of pulling back here, there are three repos that we were bouncing back and forth in this stream. There was literature, template, and curriculum. Literature did the literature meta analysis. Template is the one that you can use to make the software package and render the PDF. And then curriculum is using the local LLM to generate the course material types, audience, language, all that. Um, some key themes, generative education. I mean, education was always generative, always synthetic, always collective, always kill creative. And now that's playing out at a speed and tempo and mode that boggles the minds. How do we use AI to create useful materials and and know when not to? What does human in the loop mean? Or as Michael Garfield calls it, human on the loop. How do we maintain faculty oversight and expertise? And I'm not just talking about associate or tenure track. I mean, how do we use our attention and our faculties to be part of this co-creation lest we get sidelined and tossed aside? customization where in terms of the initial setup to the implementation to the investigation of intermediate file types, how do we customize and how do we have the foresight and the situational awareness to know what is relevant where kind of like a pre-relevance realization theory of mind theory of relevance realization quality assurance and the future of teaching.

future of teaching and learning few different discussion points then I will look in the chat to see if anybody has written any comments or questions maybe push some final updates and then call it a stream for now. So how can faculty and faculties leverage AI tools without comprising educational quality? What new skills and workflows do educators need in the age of generative AI? How do we balance efficiency gains with pedagogical depth? What are the ethical considerations in AI generated educational content? How can institutions support faculty in adopting these tools? So, let's see where we're going to get today. We're going to get two. We got a uh 450 paper literature meta analysis on Let's regenerate the paper. with 10 modules, 10 sessions on tree grafting, all just spiraling out of a 4 billion parameter language model. So not even using internet search on that, not even using a larger model, small language model generating this course and then in the context of a well ststructured template repo, we are getting a nicely formatted paper that we will look in just a second. Okay. And we'll we'll let the LLM run here. So here it's going to write up uh some peerreview comments and summaries of the paper and also translate it into Russian, Chinese, Hindi. So from about an hour ago, Tucker's suggestion that we study tree grafting, we now have a 55page paper, the tree grafting science and practice, biological mechanisms, computational analysis, and decision support framework. A comprehensive synthesis of 4,000 years of agricultural knowledge. Wow. Tree grafting represents one of humanity's oldest and most sophisticated agricultural techniques with documented use spanning over 4,000 years across diverse civilizations. This comprehensive transdiciplinary review synthesizes biological mechanisms, historical development, technical methodologies, agricultural applications, economic impacts and cultural significance of tree grafting and everything is all uh nicely formatted in this PDF. So we can click around and look at the uh the whip and tongue and and the point of the triple juxtiposition with curriculum literature and template is like you can imagine being in the special operations research team or the infrastructure division or knowledge engineer at the school of tomorrow or whatever your institution is going to be. And you could imagine going back and forth in series or in parallel doing research, expanding that meta analysis to include blogs and social media and and all these other kinds of sources. So going out there, foraging, epistemic foraging for what is out there, bringing it in, curating, representing it, building awareness, communicating that using that information and that grounding that you went out and forged for plus the semantics of existing information in your language model to develop curriculum and research. So we could bring that bib file from the literature and we could drop the bib file and the summaries into template and use that to constrain which citations or which topics we explore. And then whether you do this before or after your paper is out, we we can make curriculum around material. So it expands the prism of science communication and participation and inclusion when we can develop custom targeted starting point curriculum for any topic. Super fun. right now it's doing the peer review and then it will do the translation on the paper. We'll take a look at that, but I think people kind of get where that would go. So, that gets us through here and I'm going to copy out any points from chat. Okay. Thank you, Tucker, for the suggestions. Alex, good to see you. Thank you. Okay, I'll copy out some questions from a few if I recognize the names and icons appropriately. Very, very skilled teachers and learners. All right. So, Buzz Toot writes, "How does Daniel see tool adoption changing where we think and not just wondering about how we think?" Well, as as with everything within this whimsical who's on first style genre, which hasn't come up in a in a in a few streams, I know. um it's just just a starting point and a work in progress. But when I get a first take on that, I go from well zooming out from how do we think is where, why, and when are we thinking just like it's like saying well what's next from the triple threat that again that depends on where you are in the basketball court and game season and all that. So where we think I think that could be ranging from the most deflationary spatialized like people could just pop off a voice memo while they're on public transit or write uh on a piece of paper or in their phone as they're waking up. And that thinking can be part of an incrementally expanding active umbrella of both breadth and depth. So where we think it's going to be everywhere and and I don't even think everywhere and nowhere. I think everywhere. What is thinking? Will these tools put us further out into the world or will it allow us to dive deeper into the epistemological world of refinement or both? How does that map versus territory differential now play out? Yeah. So I I to the first part of the question I hope it would be the possibility for both and when what's called for is to go further out and to do edge computing where none have gone before. Great. if there's a vector where that boots on the ground in the world throwness is refinement amazing. If there are uh divergence kind of like a golden compass where one of the arms is going further out into practice and in into the world out there and the other arm of that golden compass is diving deeper into the world of abstraction, sophistication, refinement, interior epistemology, private language. if sort of the um I'm working with a uh John Dunn sort of compass metaphor here where here it's two lovers who are the arms of that compass and they circle around each other. In that metaphor, I'm thinking, well, maybe those two points are one and the same. Maybe one of them stays fixed. Maybe where somebody is in the world stays and they're able to spin circles and have a epistemological perimeter around them. Or maybe somebody stays grounded on the epistemological pivot and they trace a circumference, a great circle out in the world. or maybe they walk kind of one moving forward than the other, zigzagging all the way. So, I think both of these things could happen, especially if you have the when not to use it or when to use it question. Just the question of when at all. So, it's not like we're locked into this cockpit and we must use this AI. We must take the first response that it comes off the presses. This is like when we can have both eyes opens, then we can do any number of those pivot maneuvers. How does the map versus territory play out? Yeah, that's very interesting. I think part of the the fascinating aspects of cyber physical is restricting to just within this niche. Let's look at the peer review for our uh paper. So, I'm going to update tree. So, we're going to go to template. Ironically, branch grafting output llm. Okay, so here's the peer review. This is going to get to the the map territory question. Okay, got translations quality review. Okay. Some sections could be elaborated, specific issues, recommendations. Here's a Hindi translation, Russian translation, Chinese translation. probably are better ways to do translations, but that's a situation when we're playing around with this where the territory of the repo and and the map become their own sort of singularity. And it's one thing to be doing just symbolic manipulation within talking about biology labs and what could be, but it's going to collapse to actuality, including the open-endedness of actuality with what you actually have at hand. So, are we talking about the map of the territory within the repo, within the digital, or are we talking about real maps, real territories? Um, okay. Kirby wrote, "Literature analysis seem to pull in large amounts of paper outside the focus of tree grafting, getting into heart surgery big time and some algebra. A more loquacious prompt would probably fix that. Absolutely. Having well first you could just curate the list of papers or you could have some secondary thresholds. Like it's very common in meta analysis to say we did a keyword search we got a thousand papers and then we restricted to only within the last 20 years and only of this type and then we had 45 papers and then we did this and we did full data analysis and we had 26 papers and now we're going to make this meta analysis based upon 26 papers whatever it is. So you could do positive or negative terms or otherwise manually curate which papers you throw in there. A teacher would know how to graft in the field. I.e. curriculum generation does not replace skills doing heart surgery. No substitute for practice makes perfect. Yeah. Whether it's whether it's the the horiculture or the artery grafting which is still tree grafting or the algebra. It's just a topic that we always explore these multiple overloaded namespaces and uh and uh all the delights of syntax and semantics. So I think that is good for now. We have tree everything was updated the paper is updated on the tree branch of template. But I'm going to flip back over to main curriculum is updated with everything that happened. Ending the stream. So, hope that was useful or funny or interesting or all the above for those who enjoy. Certainly, it's been fun over the last few days and weeks and months and years explore all these different corners and and what is now possible and of course what happens next. So thank you all. Till next time.